

# Maximum Likelihood Estimation (MLE)

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The likelihood function is defined as:

$$L(\theta) = \prod_{i=1}^n f(x_i | \theta)$$

The log-likelihood is:

$$\ell(\theta) = \sum_{i=1}^n \log f(x_i | \theta)$$

The Maximum Likelihood Estimator is:

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta)$$

Lets see how we can implement this...

```
# Simulate some data
set.seed(123)
data <- rnorm(100, mean = 5, sd = 2)

# log-likelihood function
log_likelihood <- function(params) {
  mu <- params[1]
  sigma <- params[2]

  if (sigma <= 0) return(-Inf)

  ll <- sum(dnorm(data, mean = mu, sd = sigma, log = TRUE))
  return(ll)
}

# Optimisation (MLE)
result <- optim(
  par = c(0, 1), # Initial guesses
  fn = function(p) -log_likelihood(p), # Minimize negative log-likelihood
  method = "L-BFGS-B",
  lower = c(-Inf, 0.0001)
)

result$par

## [1] 5.180817 1.816489
```

```
# Extract MLE estimates
mu_hat <- result$par[1]
sigma_hat <- result$par[2]
```

```
mu_hat
```

```
## [1] 5.180817
```

```
sigma_hat
```

```
## [1] 1.816489
```

```
# MLE estimates for normal distribution
mean(data)
```

```
## [1] 5.180812
```

```
sd(data)
```

```
## [1] 1.825632
```

$$f(x | \lambda) = \lambda e^{-\lambda x}$$

```
exp_loglik <- function(lambda) {
  if (lambda <= 0) return(-Inf)

  sum(dexp(data, rate = lambda, log = TRUE))
}
```

```
# MLE estimation
result_exp <- optim(
  par = 1,
  fn = function(l) -exp_loglik(l),
  method = "L-BFGS-B",
  lower = 0.0001
)
```

```
lambda_hat <- result_exp$par
lambda_hat
```

```
## [1] 0.1930217
```

$$\hat{\lambda} = \frac{1}{\bar{x}}$$

```
# Theoretical MLE check
1 / mean(data)
```

```
## [1] 0.1930199
```